Tryptophan metabolism Analysis Results

Client	HEM pharma
Company | Institution	HEM pharma
Report No.	HEM-ARR-204
Reported Date	2026-03-31
Number of Samples	34

경기 수원시 영통구 창룡대로 256번길 50
광교플렉스데시앙 716호 (16229)

Tel. 031-214-9421
Fax. 070-8680-9311

CONTENTS

Tryptophan

Appendix

00. Sample information

Sample type	Feces
Number of Samples	34
Analysis target	Tryptophan metabolism
Analytical instrument	LC-MS/MS

» LC-MS/MS : Liquid Chromatography Triple Quadrupole Mass Spectrometry

01. Sample preparation

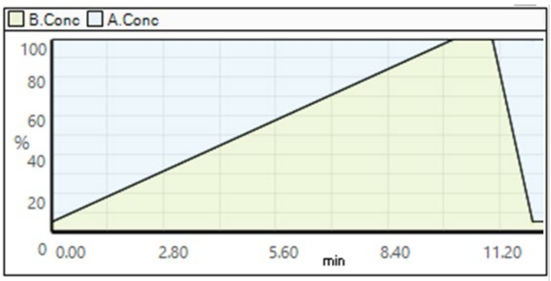
No.	Sample ID	Weight (mg)	50% Methanol (mL)
1	FD-599	48.75	0.975
2	FD-602	49.30	0.986
3	FD-605	48.64	0.973
4	FD-606	49.08	0.982
5	FD-607	49.49	0.990
6	FD-610	49.85	0.997
7	FD-616	48.36	0.967
8	FD-622	49.52	0.990
9	FD-624	48.35	0.967
10	FD-629	48.63	0.973
11	FD-642	49.58	0.992
12	FD-646	48.92	0.978
13	FD-648	48.61	0.972
14	FD-649	49.49	0.990
15	FD-706	49.40	0.988
16	FD-737	49.61	0.992
17	FD-762	48.49	0.970
18	FD-765	48.93	0.979
19	FD-775	48.55	0.971
20	FD-790	49.86	0.997
21	FD-791	49.08	0.982
22	FD-803	49.70	0.994
23	FD-868	48.91	0.978
24	FD-916	48.80	0.976
25	FD-920	50.00	1.000
26	FD-921	48.76	0.975
27	FD-926	48.95	0.979
28	FD-945	48.23	0.965
29	FD-952	49.49	0.990
30	FD-1016	48.23	0.965
31	FD-1033	49.58	0.992
32	FD-1046	49.15	0.983
33	FD-1078	49.12	0.982
34	FD-1079	48.64	0.973

» Weight (mg) : 분석에 사용한 시료의 양

» Methanol (mL) : 분석에 사용한 50% 메탄올의 양

02. Instrument information

Analytical instrument	Shimadzu LCMS-8050
Column information	ACQUITY UPLC HSS T3 Column (1.8 µm, 2.1 mm X 150 mm)
Run time	12 min / sample
Column temperature	40 °C
Mobile phase	(A) Water with 0.1% (v/v) Formic acid (B) Acetonitrile with 0.1% (v/v) Formic acid
Flow rate	0.3 mL / min
Injection volume	1 µL



No.	Time	B.Conc
1	0.00	5
2	10.00	100
3	11.00	100
4	12.00	5

Figure 1. Gradient system used in the LC-MS/MS method

» **(Fig. 1)** 분석 시간(Run time)에 따른 이동상(Mobile phase) 변화

03. Metabolite analysis results

Sample information		Concentration (μmol/g)																	
No.	Sample ID	APAP	AP	AA	DA	GABA	GLU	HA	IND	IAM	IAA	ILA	IPA	KYNA	NA	PA	TA	TRP	XA
1	FD-599	0.000	0.003	0.000	0.000	0.000	3.165	0.000	0.532	0.000	0.029	0.000	0.002	0.000	0.083	0.002	0.125	0.091	0.001
2	FD-602	0.000	0.000	0.000	0.001	0.653	0.434	0.000	0.278	0.000	0.008	0.008	0.010	0.000	0.061	0.000	0.001	0.527	0.005
3	FD-605	0.000	0.000	0.000	0.000	0.000	5.363	0.000	0.569	0.000	0.008	0.002	0.001	0.000	0.072	0.012	0.004	0.946	0.004
4	FD-606	0.000	0.003	0.000	0.000	0.000	3.044	0.000	0.524	0.000	0.002	0.000	0.000	0.001	0.018	0.001	0.233	0.000	0.001
5	FD-607	0.245	0.000	0.000	0.000	0.000	8.588	0.000	2.076	0.000	0.058	0.000	0.001	0.004	0.215	0.000	0.000	1.295	0.008
6	FD-610	0.000	0.001	0.000	0.000	0.181	3.545	0.000	0.520	0.000	0.003	0.000	0.002	0.000	0.100	0.000	0.000	0.619	0.004
7	FD-616	0.000	0.001	0.000	0.000	0.074	7.819	0.000	0.777	0.000	0.060	0.000	0.004	0.002	0.221	0.000	0.000	0.652	0.004
8	FD-622	0.000	0.001	0.000	0.000	0.000	10.419	0.000	0.830	0.000	0.119	0.004	0.007	0.003	0.331	0.003	0.000	0.275	0.005
9	FD-624	0.000	0.001	0.000	0.000	0.147	7.544	0.000	1.914	0.000	0.023	0.000	0.007	0.003	0.155	0.000	0.000	0.987	0.007
10	FD-629	0.000	0.001	0.000	0.000	0.298	1.237	0.000	0.750	0.000	0.000	0.000	0.004	0.001	0.090	0.000	0.002	0.467	0.003
11	FD-642	0.000	0.000	0.000	0.000	0.896	6.061	0.000	1.246	0.000	0.012	0.005	0.002	0.000	0.161	0.000	0.214	1.497	0.009
12	FD-646	0.000	0.001	0.000	0.000	3.312	0.466	0.000	0.226	0.000	0.003	0.000	0.004	0.000	0.071	0.000	0.163	1.716	0.008
13	FD-648	0.002	0.000	0.000	0.000	0.000	1.358	0.000	0.327	0.000	0.019	0.000	0.001	0.002	0.029	0.000	0.000	0.165	0.002
14	FD-649	0.000	0.002	0.000	0.000	0.000	3.567	0.000	0.893	0.000	0.000	0.000	0.001	0.000	0.065	0.000	0.000	0.312	0.002
15	FD-706	0.000	0.002	0.000	0.000	0.000	6.202	0.000	1.272	0.000	0.016	0.005	0.001	0.000	0.112	0.000	0.006	0.298	0.011
16	FD-737	0.000	0.001	0.000	0.000	0.000	5.463	0.000	0.559	0.000	0.007	0.001	0.008	0.003	0.285	0.000	0.000	0.445	0.005
17	FD-762	0.000	0.000	0.000	0.000	0.311	3.911	0.000	0.469	0.000	0.063	0.000	0.002	0.001	0.091	0.000	0.000	0.634	0.003
18	FD-765	0.000	0.001	0.000	0.000	0.000	3.669	0.000	1.628	0.000	0.012	0.000	0.007	0.003	0.020	0.000	0.002	1.318	0.007
19	FD-775	0.000	0.000	0.000	0.000	0.324	6.015	0.000	0.815	0.000	0.054	0.000	0.003	0.000	0.133	0.000	0.005	0.863	0.006
20	FD-790	0.000	0.000	0.000	0.000	0.044	2.895	0.000	0.244	0.000	0.000	0.000	0.001	0.000	0.074	0.000	0.053	0.497	0.003
21	FD-791	0.000	0.000	0.000	0.000	0.000	7.103	0.000	1.371	0.000	0.022	0.003	0.004	0.001	0.231	0.000	0.000	0.716	0.004
22	FD-803	0.000	0.000	0.000	0.000	0.060	10.412	0.000	0.576	0.000	0.015	0.003	0.002	0.002	0.091	0.003	0.000	0.723	0.005
23	FD-868	0.000	0.002	0.000	0.001	0.681	4.404	0.000	1.078	0.000	0.002	0.001	0.004	0.001	0.135	0.000	0.008	0.297	0.002
24	FD-916	0.502	0.000	0.000	0.000	0.483	5.261	0.000	0.596	0.000	0.013	0.000	0.004	0.003	0.194	0.000	0.003	1.471	0.011
25	FD-920	0.000	0.000	0.004	0.000	0.000	2.993	0.000	0.352	0.000	0.009	0.000	0.003	0.000	0.009	0.000	0.000	0.804	0.004
26	FD-921	0.000	0.001	0.000	0.000	0.059	4.338	0.000	0.798	0.000	0.000	0.000	0.003	0.001	0.084	0.000	0.000	0.872	0.005
27	FD-926	0.000	0.000	0.000	0.000	0.000	2.652	0.000	0.501	0.000	0.104	0.000	0.002	0.002	0.047	0.000	0.000	0.375	0.002
28	FD-945	0.000	0.000	0.000	0.001	0.093	4.412	0.000	0.259	0.000	0.002	0.000	0.003	0.001	0.116	0.000	0.000	0.514	0.004

29	FD-952	0.000	0.000	0.000	0.000	0.000	4.350	0.000	0.578	0.000	0.000	0.007	0.001	0.000	0.037	0.002	0.000	0.462	0.003
30	FD-1016	0.000	0.000	0.000	0.000	0.000	3.225	0.000	0.282	0.000	0.000	0.000	0.000	0.002	0.026	0.000	0.000	0.427	0.003
31	FD-1033	0.000	0.000	0.000	0.000	0.000	5.653	0.000	1.059	0.000	0.005	0.001	0.002	0.005	0.088	0.000	0.000	0.490	0.004
32	FD-1046	0.000	0.001	0.000	0.000	0.077	8.208	0.000	0.818	0.000	0.015	0.002	0.002	0.003	0.174	0.000	0.000	0.539	0.004
33	FD-1078	0.003	0.000	0.000	0.000	1.337	7.532	0.000	0.713	0.000	0.003	0.002	0.009	0.003	0.116	0.000	0.000	1.006	0.008
34	FD-1079	0.000	0.000	0.000	0.000	0.000	5.473	0.000	0.182	0.000	0.000	0.000	0.008	0.001	0.121	0.000	0.000	0.705	0.005

» Abbreviations : APAP, Acetaminophen; AP, Aminophenol; AA, Anthranilic acid; DA, Dopamine; GABA, gamma-Aminobutyric acid; GLU, Glutamic acid; HA, Histamine; IND, Indole; IAM, Indole-acetamide; IAA, Indole-acetic acid; ILA, Indole-lactic acid; IPA, Indolepropionic acid; KYNA, Kynurenic acid; NA, Nicotinic acid; PA, Picolinic acid; TA, Tryptamine; TRP, Tryptophan; XA, Xanthurenic acid

» Concentration (μmol/g) : 분변 g 당 검출된 각 담즙산의 농도(μmol)

Appendix

01. Sample preparation and Data analysis

Sample → Metabolite Extraction → Instrument Analysis → Data processing

- **Sample** : 시료는 입고 즉시 시료의 변질을 최소화 하기 위해 실험 전까지 deep-freezer(-80 ℃)에 보관합니다.

- **Metabolite extraction** : 혈청 볼륨에 4배에 해당하는 100% 메탄올을 첨가하여 충분히 볼텍싱 후 다음 원심분리를 합니다. 상층액 필터 후, 96 well plate에 옮깁니다

- **Instrument analysis** : Instrument information 에 따라 LC-MS/MS분석을 수행합니다.

- **Data processing** : 혈청에서 검출된 대사체에 대한 calibration curve를 구축하고, 상관계수 r > 0.999의 값을 확인하며 해당 curve를 기반으로 분변 내 각 대사체에 대한 정량 값을 산출합니다.

Disclaimer